STM Paper Implementation Guidelines

Paper title: *STM: Spectrogram Transformer Model for Underwater Acoustic Target Recognition*

Paper link to download: https://www.mdpi.com/2077-1312/10/10/1428

Max member in a group: 3

Dataset Link to download: In this there is a readme file and a excel file (meta data) containing name of vessels to categories in different classes.

https://gusindia01-my.sharepoint.com/:f:/g/personal/seetaram_maurya_ddn_upes_ac_in/IgBQx7Wi_r2FSZOmLNZIJx4AARs30co3RobDf0L8j5Mor8c?e=eUhYfV

Part 1 — Paper Understanding

Deadline: 15 October

Students have to:

1. Read and understand the complete paper.
2. Understand the **ShipsEar dataset** and 5 target classes.
3. Understand how the audio is converted into **5-second samples**.
4. Understand **Dataset A and Dataset B** and their splitting methods.
5. Understand the three features:
    - STFT
    - FBank
    - MFCC
6. Understand the **CNN, ResNet18 and STM** architectures.
7. Understand STM:
    - Spectrogram → patches
    - Patch embedding
    - Positional embedding

    - Transformer Encoder
    - MLP classifier
8. Note all important parameters from the paper.
9. Prepare the Google Colab notebook structure (Colab notebook skeleton).

Part 2 — Dataset and Feature Extraction

Deadline: 30 October

Students have to:

1. Download the **ShipsEar dataset** from the provided link above.
2. Convert audio into **5-second samples**.
3. Remove/add unusable samples if any.
4. Create and save **Dataset A** using the paper's random split.
5. Create and save **Dataset B** using the paper's chronological split.
6. Implement below feature extraction:
    - STFT
    - FBank
    - MFCC
7. Generate and save feature representations.
8. Implement the modified **ResNet18**.
9. Train ResNet18 using the extracted features.
10. Reproduce the experiment required for **Table 2**.
11. Compare:

- Paper result
- Student result

Output: Dataset A and B + features + ResNet18 + Table 2 reproduction.

Part 3 — Complete paper Implementation

Deadline: 15 November

Students have to on both data A and B:

1. Implement **CNN baseline**.
2. Implement **ResNet18 baseline**.
3. Implement **STM**.
4. Convert spectrogram into **16×16 patches with overlap 6**.
5. Implement patch embedding and positional embedding.
6. Implement the **12-layer, 12-head Transformer Encoder**.
7. Add the MLP classification head.
8. Train STM using the paper's training settings.
9. Implement the required pretrained experiments where possible:
    - ImageNet
    - AudioSet
10. Implement time/frequency masking augmentation.
11. Train and test the models on **Dataset A and Dataset B**.
12. Generate:

- Accuracy
- Confusion matrix
- Training/validation curves

13. Compare reproduced results with the paper.

Output: Complete working STM implementation + experimental results.

Part 4 — Final Submission and Presentation

Deadline: 20 November

Students have to:

1. Clean and organize the complete Colab notebook.
2. Ensure the notebook runs from beginning to end.
3. Save trained model/checkpoints.
4. Prepare final result tables.
5. Compare **paper vs reproduced results**.
6. Explain differences in results.
7. Prepare a short presentation/report covering:
    - Problem
    - Dataset
    - Feature extraction
    - STM architecture
    - Implementation
    - Results
    - Comparison
    - Conclusion

Submission Instructions

Students have to **implement/reproduce the given research paper in a Jupyter Notebook using Google Colab**.

Requirements

1. Create a Google Drive folder named:
   GroupName_sapIDs
2. Perform the complete implementation in a **Jupyter Notebook (.ipynb)** using Google Colab.
3. The main .ipynb notebook should include:
    - Dataset preparation
    - Dataset A and Dataset B creation

- STFT, FBank and MFCC feature extraction
- CNN and ResNet18 implementation
- STM implementation
- Results and comparison with the paper

4. Save **all required files, datasets/data paths, code, trained models, results, figures, and documentation** inside the Google Drive folder.
5. Maintain an organized folder structure, for example:

Students must submit the **Google Drive folder link** containing all files/folders required to run and verify the implementation. You have to ensure that the folder has the required access permission and contains the **complete implementation and supporting files**.

Submission link: Project Submission – Fill out form